

Convex Analysis and Linear Logic: A Research Program

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Abstract

Convex analysis and linear logic exhibit strikingly similar patterns of duality and algebraic operations despite having been developed for entirely different purposes. This research program explores the hypothesis that these parallels arise from a common underlying geometric structure. We propose *preepi*, an arbitrary subset of an extended space, as a primitive geometric object whose regular instances may represent closed convex functions, while its intrinsic algebra may provide a model of a fragment of linear logic. Because *preepis* are more general than epigraphs of convex functions, this algebra may also carry structures with no immediate counterpart in classical convex analysis. Rather than presenting established results, this document formulates the basic objects, operations, conjectures, and milestones needed to investigate these possibilities. If successful, it would place convex analysis within a broader geometric framework and reveal a structural connection with linear logic.

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1 Introduction

This program began from a simple observation. In convex analysis, the Fenchel conjugate f^* is an order reversing involution on closed convex functions, epigraphs combine under closed pointwise addition and under closed infimal convolution, and both operations are dual to each other through conjugation. In linear logic, linear negation is an order reversing involution on formulas, and the multiplicative connectives $\otimes$ and $\wp$ are dual to each other through that negation. We noticed that these two patterns match closely enough, in shape if not yet in substance, that we wanted to know whether the match is accidental or structural.

To investigate the apparent structural parallels between convex analysis and linear logic, we propose an intermediate geometric object, called a *preepi*, defined as an arbitrary subset of $X \times$

$[-\infty, +\infty]$ without any assumption of convexity or epigraph structure. The central hypothesis of this research program is that preepi carries an intrinsic algebraic structure independent of either theory, while closed convex functions and linear logic arise as two distinct representations of that common structure. This separation is intended to distinguish the underlying geometry from its analytic and logical interpretations, and to provide a common framework in which the relationship between the two fields can be studied systematically.

Multiplicative additive linear logic, MALL, is the fragment of Girard's linear logic obtained by dropping the exponential modalities $!$ and $?$ and any quantifiers. It retains the four binary connectives, their units, and linear negation. Full linear logic recovers resource duplication and discarding through $!$, $?$ on top of this fragment. The claims in this program are stated for MALL. A candidate pair of operators intended to play the role of $!$, $?$ is proposed in Section 2.4, together with the open question, deferred to Milestone III, of whether that candidate actually satisfies the laws expected of exponentials.

We follow the algebraically oriented notation of Paoli, writing $\wedge, \vee, \otimes, \oplus$ for the connectives with, plus, tensor, and par, respectively, rather than Girard's notation $\&, \oplus, \otimes, \wp$, as this aligns the logical symbols directly with the lattice and monoidal operations on preepis.

We propose the following correspondence as the organizing table of the program, restricted to the four MALL connectives, negation, and their units. The residual and the exponential candidate are proposed separately in Section 2.4 and are not included in this table.

MALL, Paoli notation	Preepi operation	Function level mirror
pole, pairing	$\beta(x, y) := \langle x, y \rangle$	—
$\neg$, negation	$E \mapsto E^*$, Fenchel polar	$f \mapsto f^*$
fact, $A = \neg\neg A$	$E = E^{**}$, i.e. $E \in \overline{X}$	$f = f^{**}$
$\vdash$, provability order	$E \subseteq F$	$f \geq g$
$\wedge$, with	$E \wedge F := E \cap F$	$\max(f, g)$
$\vee$, plus	$E \vee F := (E \cup F)^{**}$	$(\min(f, g))^{**}$
$\otimes$, tensor	$E \otimes F := (E +_{\text{fib}} F)^{**}$	$f + g$
$\oplus$, par	$E \oplus F := (E^* \otimes F^*)^*$	$\text{cl}(f \square g)$
unit of $\otimes$	$1_{\otimes} := X \times [0, +\infty]$	$\text{cl}(0)$
unit of $\oplus$	$1_{\oplus}$, see Claim 9	$\text{cl}(\delta_{\{0\}})$
unit of $\wedge$	$1_{\wedge} := X \times [-\infty, +\infty]$	$f \equiv -\infty$
unit of $\vee$	$1_{\vee} := X \times \{+\infty\}$	$f \equiv +\infty$

The correspondence table above should be regarded as a working hypothesis rather than a finalized dictionary. While some entries are supported by well established results from convex analysis, others are proposed as natural modeling choices whose validity remains open. In particular, the identification of $\otimes$ and $\oplus$, the interpretation of the order $\vdash$, and the relationship between the order on preepis and the pointwise order on functions are all tentative and may require revision. Likewise, the residual $\multimap$ is proposed as a derived operation rather than an interpretation of the order itself.

The roadmap consists of three milestones, one per stage. Milestone I concerns only the intrinsic algebra of preepi, without reference to convex functions or to logic. Milestone II connects that algebra to closed convex functions. Milestone III connects it to linear logic, both to MALL as claimed throughout the document and, as a further direction, to the exponential candidate beyond MALL. Each milestone can stand or fall independently of the ones after it.

2 The components

Every item below is a claim, a proposal for how the corresponding object should be modeled. Motivation from the function level picture is given alongside each one.

2.1 Preepi, closures, and regepi

Claim 1 (Preepi). *Fix*

$$X := \mathbb{R}^n, \quad \overline{\mathbb{R}} := [-\infty, +\infty], \quad \tilde{X} := X \times \overline{\mathbb{R}}.$$

A preepi is an arbitrary subset $E \subseteq \tilde{X}$.

Motivation. A point $(x, r) \in \tilde{X}$ records a location together with a height. Both $+\infty$ and $-\infty$ are retained so that a preepi can model a total function $f : X \rightarrow \overline{\mathbb{R}}$, with $+\infty$ encoding points outside the effective domain and $-\infty$ a degenerate, unbounded below value, and so that the two lattice units introduced later both have a genuine function level mirror. Retaining both infinities has one cost: sums of the form $r + s$ can involve $(+\infty) + (-\infty)$. Throughout this document this sum is fixed to equal $+\infty$, the Rockafellar convention; this is a choice, not a derived fact, and it is what makes the polar of Section 2.2 well defined on all of $\tilde{X}$.

Claim 2 (Closures). *For $E \subseteq \tilde{X}$ and $x \in X$, write $E_x := \{r \in \overline{\mathbb{R}} : (x, r) \in E\}$. Define*

$$\begin{aligned} U(E) &:= \{(x, r') : \exists r \leq r', (x, r) \in E\} && (\text{upward closure}) \\ F(E) &:= \{(x, r) : r \in \text{cl}(E_x)\} && (\text{fiber closure}) \\ C(E) &:= \text{conv}(E) && (\text{convex closure}) \\ T(E) &:= \text{cl}(E) && (\text{topological closure}) \end{aligned}$$

Motivation. Each operator repairs one property separating a general preepi from the extended epigraph of a proper, convex, lower semicontinuous function: upwardness in each fiber, fiber closedness, global convexity, global closedness.

Why exactly four, and not three or five. The four properties above are proposed as a candidate decomposition of the geometric features that distinguish an arbitrary preepi from an extended epigraph. They repair, respectively, the order structure of each fiber (upwardness), the topology of each fiber (fiber closedness), the global algebraic structure (convexity), and the global topology (closedness). Whether these four properties are indeed independent, whether they suffice to characterize extended epigraphs, and whether additional intrinsic conditions are needed remain open questions to be addressed in Milestone I.

Claim 3 (Regepi). *A preepi E is a regepi if it is regular in the sense above, that is, if $U(E) = F(E) = C(E) = T(E) = E$. Write $\overline{X} := \{E \subseteq \tilde{X} : E \text{ is a regepi}\}$.*

Motivation. A regepi is proposed as exactly the class of preepis arising as extended epigraphs of proper, convex, lower semicontinuous functions. An additional intrinsic condition will likely be needed to characterize proper functions, and is left open.

Claim 4 (Reflector property). *Each of U, F, C, T is a reflector from the poset of all preepis, ordered by inclusion, onto the sub-poset of preepis fixed by that operator: for every preepi E and every c -closed preepi A , with c one of U, F, C, T ,*

$$E \subseteq A \iff c(E) \subseteq A.$$

Motivation. This is an equivalent, universal property style restatement of Claim 2: it says $c(E)$ is the smallest c -closed preepi containing E , for each of the four properties in turn, rather than introducing any new content. It is recorded separately mainly because, combined with whether the four closures commute in every order, it frames the composite map $E \mapsto R(E) := T(C(F(U(E))))$ used to regularize an arbitrary preepi into a regepi as a specific, order sensitive composition of four reflectors rather than an arbitrary formula, a fact needed in Milestone I.

2.2 Order and polarity

Claim 5 (Order). *Preepis are ordered by inclusion: $E \subseteq F$.*

Claim 6 (Polar). *For $E \subseteq \tilde{X}$,*

$$E^* := \{(y, s) \in \tilde{X} : \langle x, y \rangle \leq r + s \quad \forall (x, r) \in E\},$$

with a fixed Euclidean inner product on X , and $r + s$ computed by the Rockafellar convention fixed above.

Motivation. This is the geometric form of the Fenchel conjugate $f^*(y) := \sup_x \{\langle x, y \rangle - f(x)\}$, proposed so that $E = \text{epi}(f)$ implies $E^* = \text{epi}(f^*)$. Because $s = +\infty$ always satisfies the defining inequality, including against $r = -\infty$ by the Rockafellar convention, $E^* \neq \emptyset$ for every preepi E .

What this raises for regepi. Applying the polar twice gives E^{**} , and $E \subseteq E^{**}$ holds for an arbitrary preepi by the defining inequality alone. Whether $E^{**} = E$ exactly when E is a regepi, and what happens when E is a regepi but $E^{**} \neq E$, is not settled by the definitions above and is taken up as one of the central questions of Milestone I.

2.3 Four operators and four units

The four operators below, $\wedge, \vee, \otimes, \oplus$, are the main objects of study in this program. Two further constructions, fiber sum and Minkowski sum, are also defined here, but only as raw geometric ingredients from which $\otimes$ and $\oplus$ are built; they are not themselves claimed to carry the algebraic structure that the induced operators are claimed to carry.

Claim 7 (Geometric ingredients). *For $E, F \subseteq \tilde{X}$,*

$$E +_{\text{fib}} F := \{(x, r+s) : (x, r) \in E, (x, s) \in F\}, \quad E +_{\text{M}} F := \{(x_1+x_2, r_1+r_2) : (x_1, r_1) \in E, (x_2, r_2) \in F\}.$$

Motivation. Fiber sum mirrors pointwise addition of functions, $\text{epi}(f) +_{\text{fib}} \text{epi}(g) \leftrightarrow \text{epi}(f+g)$, with no closure needed at the function level. Minkowski sum mirrors the numerator of infimal convolution, $\text{epi}(f) +_{\text{M}} \text{epi}(g) \leftrightarrow \text{epi}(f \square g)$ before closure.

Claim 8 (Four operators). *For $E, F \in \overline{X}$,*

$$E \wedge F := E \cap F, \quad E \vee F := (E \cup F)^{**}, \quad E \otimes F := (E +_{\text{fib}} F)^{**}, \quad E \oplus F := (E^* \otimes F^*)^*.$$

Motivation. $\wedge, \vee$ mirror $\max(f, g)$ and the bipolar closure of $\min(f, g)$. $\otimes$ mirrors closed pointwise addition, $f+g$. $\oplus$ is proposed not directly from Minkowski sum but through De Morgan duality from $\otimes$, mirroring the identity $(f \square g)^* = f^* + g^*$, which holds unconditionally at the function level unlike its converse, so that $\oplus$ is automatically valued in $\overline{X}$ without a separate argument. The candidate identity $E \oplus F = (E +_{\text{M}} F)^{**}$, connecting back to the geometric ingredient of Claim 7, is not claimed here and is taken up in Milestone II.

Claim 9 (Four units).

$$1_{\wedge} := X \times \overline{\mathbb{R}}, \quad 1_{\vee} := X \times \{+\infty\}, \quad 1_{\otimes} := X \times [0, +\infty], \quad 1_{\oplus} := (\{0\} \times [0, +\infty]) \cup ((X \setminus \{0\}) \times \{+\infty\}).$$

Motivation. $1_{\wedge}, 1_{\vee}$ mirror the constant functions $f \equiv -\infty$ and $f \equiv +\infty$, both genuine total functions $X \rightarrow \overline{\mathbb{R}}$ now that $\widetilde{X} = X \times [-\infty, +\infty]$. Neither belongs to $\Gamma_0(X)$, since both violate properness, so this symmetry is a gain purely at the level of the intrinsic algebra, not an enlargement of the class of functions targeted in Milestone II. $1_{\otimes}$ mirrors the zero function, the unit of pointwise addition. $1_{\oplus}$ mirrors the indicator $\delta_{\{0\}}$, the unit of infimal convolution, occupying $X \times \{+\infty\}$ away from the origin, a shape forced by every regepi's fiber being allowed to reach $+\infty$. These four objects and their pairwise duality, $1_{\wedge}^* = 1_{\vee}$ and $1_{\otimes}^* = 1_{\oplus}$, are claimed independently of which MALL connective each is assigned to.

2.4 Residual and exponential candidate

If the proposed correspondence between preepi and linear logic is correct, then the four operations introduced in the previous subsection should not exhaust the algebraic structure of preepi. From the logical viewpoint, one naturally expects further operations corresponding to logical implication and, beyond multiplicative additive linear logic, the exponential modalities. The constructions below are proposed as candidate counterparts of these operations.

The order $\subseteq$ used throughout this document is a relation between two preepi, true or false, not itself an object. The residual defined below is, by contrast, an operation: it takes $E, F \in \overline{X}$ and produces a new preepi $E \multimap F$. The two are connected, since the residual is characterized through the order by the universal property $G \otimes E \subseteq F$ if and only if $G \subseteq E \multimap F$, but they answer different questions and should not be identified with each other.

Claim 10 (Residual). *For $E, F \in \overline{X}$,*

$$E \multimap F := (E \otimes F^*)^*.$$

Motivation. At the logic level, linear implication is not primitive but is definable as $A \multimap B := \neg A \oplus B$. The formula above is the direct translation, $E \multimap F = E^* \oplus F$, rewritten through Claim 8 in terms of $\otimes$ and $(\cdot)^*$ alone. Because $(\overline{X}, \subseteq)$ is claimed in Milestone I to be a complete lattice with $\otimes$ distributing over arbitrary joins, the abstract residual $\bigvee \{G \in \overline{X} : E \otimes G \subseteq F\}$ exists unconditionally, independently of Claim 10. Whether the two agree is taken up in Milestone II.

On an exponential candidate. Full linear logic adds two further connectives, $!$ and $?$, governing how a resource may be duplicated or discarded. Nothing in the algebra developed so far forces a particular choice of $!, ?$, and none is claimed here. One idea worth recording, carried over directly from the reading of linear logic as resource management, is that a preepi restricted to nonpositive height at a point might reasonably be treated as freely duplicable there. Concretely, for $E \in \overline{X}$, let $Z_E := \{x \in X : (x, 0) \in E\}$; a possible definition along these lines is

$$!E := (Z_E \times [0, +\infty]) \cup ((X \setminus Z_E) \times \{+\infty\}), \quad ?E := (!E^*)^*,$$

stated purely in terms of E and its zero level fiber Z_E , with no reference to a function; at the function level, when $E = \text{epi}(f)$, this would mirror the indicator of the zero sublevel set $\{x : f(x) \leq 0\}$. This is offered as a candidate to be tested, not as a claim, and is kept out of the main correspondence table in Section 2 above for that reason. Whether it, or some other construction, deserves the name exponential is left entirely to Milestone III.

3 Roadmap

Three milestones are proposed, one per stage. Milestone I never mentions convex functions or logic. Milestone II interprets the algebra of Milestone I as convex analysis. Milestone III interprets it as MALL and, as a further direction, tests the exponential candidate against full linear logic. Each milestone can stand or fall independently of the ones after it.

Milestone I. Intrinsic algebra of preepi

The first part of this milestone establishes Claims 2 through 4 as a self contained algebraic package: extensivity, $E \subseteq E^{**}$, antitonicity, idempotence of the double polar, the regularizing property that E^* is always upward, fiber closed, convex, and closed for arbitrary E , closure invariance, $E^* = U(E)^* = F(E)^* = C(E)^* = T(E)^*$, and the relation, raised but not settled in Section 2.2, between $E^{**} = E$ and E being a regepi, together with the general formula for E^{**} on an arbitrary preepi. Also included are the reflector property of Claim 4, whether the four closures commute with each other and in which order the composite regularization map is well defined, and the claim that exactly four closure properties are necessary and sufficient to characterize an extended epigraph.

A further part of Milestone I concerns Claims 8 and 9: $(\overline{X}, \subseteq)$ as a complete lattice under $\wedge, \vee$, and $\otimes, \oplus$ as commutative monoids with the stated units, monotone and distributive over arbitrary joins or meets as appropriate, together with additive and multiplicative De Morgan laws and duality of the four units.

A final part concerns the residual and the exponential candidate of Section 2.4. The residual is evaluated by purely algebraic criteria, stated without reference to logic: whether the formula of Claim 10 behaves as expected of a residuated monoid. The exponential candidate is only sketched in Section 2.4; at this stage one can still ask, as a bare equation and with no logical content attached, whether some version of $!E \otimes !E = !E$ or a similar identity holds, but attaching names such as dereliction, weakening, contraction, or a Seely law to any such identity, and treating that naming as evidence for anything, belongs entirely to Milestone III.

Milestone II. Connection to convex optimization

The first part of this milestone concerns the correspondence between $\overline{X}$ and the class $\Gamma_0(X)$ of proper, convex, lower semicontinuous functions, through $E \leftrightarrow \text{epi}(f)$, and the bipolar theorem of Milestone I read as the Fenchel Moreau fixed point theorem, $f = f^{**}$ if and only if $f \in \Gamma_0(X)$. Beyond this basic correspondence, one may ask whether it extends to a genuine bijection between $\overline{X}$ and all total functions $X \rightarrow \overline{\mathbb{R}}$ satisfying properness, convexity, and lower semicontinuity, given that the height coordinate now includes $-\infty$. This first part concerns only the correspondence of objects, not yet of operators.

A second, separate part asks whether each operator of Milestone I corresponds, under that same object level correspondence, to the function level operation proposed as its mirror in Section 2: $\wedge, \vee$ to max and closed min, $\otimes, \oplus$ to closed pointwise sum and closed infimal convolution, and the units to $-\infty, +\infty, 0, \delta_{\{0\}}$. If the first part holds but the operators do not correspond as claimed, the object level correspondence would stand while the algebraic correspondence would fail; the two parts are treated as logically separate claims for exactly that reason, and the second is never assumed from the first. Within this second part, two specific representation questions stand out. One is whether $E \oplus F = (E +_M F)^{**}$ on $\overline{X}$, that is, whether the polar first definition of Claim 8 agrees with the double polar of raw Minkowski sum from Claim 7; at the function level this is the classical qualification question for exactness of infimal convolution, for instance under a relative

interior condition on the two domains. The other is whether the formula $E \multimap F = (E \otimes F^*)^*$ of Claim 10 agrees with the abstract residual defined purely order theoretically in Section 2.4, and correspondingly whether $(f + g^*)^*$ coincides with the object naturally adjoint to closed pointwise addition at the function level.

Milestone III. Connection to linear logic

The first part of this milestone asks whether the pairing β and the polar of Claim 6 instantiate, in the technical sense of Girard, a monoid acting on $\tilde{X}$ together with a pole $\perp$, so that $E^* = \{a : \forall e \in E, e \cdot a \in \perp\}$ literally, and $\overline{X} = \mathbf{Fact}_\perp(\tilde{X})$ in the standard phase semantic sense, rather than merely an algebra that happens to resemble one. Tied to this is the question, already flagged in the introduction, of which of the two operators of Claim 8 should really be identified with $\otimes$ and which with $\oplus$: the assignment adopted in this document favors fiber sum for $\otimes$ on a resource intuition basis, while Girard’s phase semantics would favor Minkowski sum, the single ambient monoid of $\tilde{X}$, unless a different ambient structure, such as a quantaloid indexed by $x \in X$, is used instead, a structure already attempted and found non associative in earlier drafts of this program.

A second part does not assume in advance that the algebra of Milestone I must match MALL exactly, and instead asks which fragment of linear logic, if any, it actually matches: possibly MALL as claimed throughout this document, possibly a proper sub-fragment if some connective or unit fails one of its expected laws, or possibly a strictly larger fragment if the exponential candidate of Section 2.4 turns out to satisfy the laws expected of $!, ?$. Settling this requires checking, once the assignment above is fixed, whether the resulting correspondence table validates every MALL inference rule, so that the semantic value of a premise lying in the claimed facts always forces the same of the conclusion, and whether the semantic assignment is well defined by structural induction on formulas at all. It also requires testing the exponential candidate directly against dereliction, weakening, contraction, digging, and the Seely isomorphisms relating $!$ to $\wedge$ and $\otimes$; if all of these were to hold, the natural reading would be that this program’s algebra models full linear logic rather than only MALL, and the scope restriction stated in the introduction would need to be revisited.

If all three milestones were carried out and answered favorably, preepi would stand as a single geometric object from which both closed convex functions and a model of linear logic could be recovered as representations, giving the two fields a shared algebraic origin rather than a resemblance noted after the fact. Nothing in this document should be read as asserting that outcome; it is stated here only as the condition under which the observation in the introduction would be explained rather than merely recorded.